

Kelowna, Canada, May 13, 2026

Dear Editors,

We are pleased to submit our manuscript entitled “*Rising temperatures alter how and where boreal mammals move*” for consideration as a research article in *Global Change Biology*.

Climate change is rapidly reshaping ecosystems around the world. A growing body of evidence is demonstrating how widespread warming during the last century has caused many terrestrial mammals to change how and where they move, with cascading effects on fitness and community dynamics. Although research on the effects of temperature on mammalian movement behaviour is growing, most studies have focused on documenting changes in only a single species, and few have disentangled temperature effects from seasonal behaviour cycles. Consequently, it is still uncertain how mammals will adapt their movement behaviour throughout the 21st century, nor what the community-level implications of such changes might be. More work is needed on quantifying how different species respond to temperature and the consequences for ecological communities.

Our study addresses this gap by modelling the effects of hourly, proximal temperature on the movement rates and habitat selection of 434 individuals across six mammal species throughout North-Western Canada, namely: boreal and southern mountain caribou, cougars, elk, grizzly bears, mountain goats, and wolves between 1998 and 2023. We employ the flexibility and interpretability of Generalized Additive Models to show that temperature has nonlinear and complex effects on mammalian movement behaviour, but that responses vary greatly across species. We then leverage the estimates to produce quantitative predictions of how climate change will affect each species’ movement behaviour throughout the 21st century. We show that, as temperatures rise, many individuals will begin selecting against their current ranges as they adapt to changing phenology and community dynamics. Finally, we present a case study using data on boreal caribou and wolves to show how rising temperatures (and concomitant changes in habitat selection) are likely to increase encounter rates between the two species due to changes in habitat selection, but that the changes are not uniform in space, and there are regions where interactions may decrease, versus others where they will increase. Consequently, rising temperatures are likely to impact wolves’ predation of caribou and the effectiveness of wolf reduction strategies.

Our findings offer new insight into the effects of climate change on the movement behaviour of multiple boreal mammalian species, that suggest conservation efforts should account for future changes in movement behaviour as well as any cascading effects on biological communities. Anticipating changes in mammalian movement behaviour will become crucial for effectively and proactively understanding community-level responses and selecting high-quality habitat for long-term conservation. We believe the manuscript will be of strong interest to researchers working at the intersection of climate adaptation, biodiversity protection, and ecological forecasting. All code and data are openly available via GitHub (<https://github.com/QuantitativeEcologyLab/bc-mammals-temperature>) to support transparent and reproducible science. The tracking data for 82% of individuals are publicly available (elk: <https://www.datarepository.movebank.org/handle/10255/move.1069>; boreal caribou and wolves: <https://www.bcogris.ca/projects/boreal-caribou-telemetry-data/>). Location data for southern mountain caribou, cougar, grizzly bear, and mountain goat cannot be shared publicly due to conservation concerns for these at-risk populations, in accordance with data sharing agreements with the relevant provincial agencies. However, all derived data products used directly in the analyses, including the estimated instantaneous movement speeds, activity states, and habitat covariates at each fix will be archived in a public repository with a persistent DOI prior to publication, along with the code required to reproduce

these analyses (<https://github.com/QuantitativeEcologyLab/bc-mammals-temperature>). These data are sufficient for independent verification of all statistical models and results reported in the paper in support of transparent and reproducible science.

Below, we include the answers to the questions requested for submission.

1. What scientific question is addressed? How does ambient temperature affect the movement rates and habitat selection of large Nearctic mammals, and how will rising temperatures under climate change alter these behaviours through to 2100?
2. What are the key findings? Temperature drives species-specific, nonlinear changes in movement and habitat selection across six large mammal species. Projections to 2100 show how some species will move more, others less. These changes have consequences for community dynamics, including predator-prey encounter rates, illustrated by a four-fold increase in caribou-wolf co-occupancy between -20°C and 20°C.
3. Why is this work important and timely? Northern regions are experiencing accelerating warming and extreme heat events. Understanding how wildlife will respond to this change is critical, but most prior work focuses on single species in isolation. This multi-species approach provides novel insight into how differing thermal responses will reshape communities and alter predator-prey dynamics.
4. How does it fit GCB's scope? GCB publishes studies on biological responses and feedbacks to change across levels of organization. This manuscript addresses organismal-level responses to temperature and scales these responses up to community-level dynamics across multiple trophic levels. This directly fitting GCB's mandate to promote understanding of biological systems and all aspects of environmental change.
5. What biological and global change aspects does it address? This manuscript examines how rising temperatures and shifting phenology alter movement behaviour and habitat selection across a community of large mammals, with cascading consequences for predator-prey dynamics and community structure. Biological responses are projected through 2100 under four IPCC Shared Socioeconomic Pathways.
6. Three most relevant recent papers? Dickie et al. (2024, GCB) identified climate as underpinning ungulate northern range expansion into caribou habitat, with community-level implications. Lessard et al. (2025, Biodiversity and Conservation) demonstrated weather and disturbance interact to shape caribou movement. Liao et al. (2026, GCB) showed how dynamic conditions under climate change alter caribou movement.
7. Provide the name and institutional email addresses of five potential reviewers. Briana Abrahms (abrahms@uw.edu), Laura Prugh (lprugh@uw.edu), Marketa Zimova (mzimova@uwo.edu), John DeLong (jpdelong@unl.edu), and David Boukal (dboukal@prf.jcu.cz)

We thank you for considering our submission and look forward to your response.

Kind regards,

Stefano Mezzini and Prof. Michael Noonan on behalf of all co-authors